

Hands-On-Learning: Log Analysis and Attack Correlation

1. Lab Overview

In this lab, you will act as a cybercrime investigator for a small business, "Innovate Solutions Inc." The company's payroll manager, Bob, reported that he received a strange email and clicked a link, though he insists he didn't enter any credentials. A day later, an unauthorized login to the company's payroll portal was detected. Your mission is to analyze a small set of provided evidence—the suspicious email's header and a snippet from the company's firewall log—to piece together what happened.

2. Learning Objectives

- Analyze an email header to identify signs of a phishing attack.
 - Analyze a firewall log to identify suspicious network activity.
 - Interpret a simplified cryptocurrency transaction log.
 - Correlate evidence from multiple sources to build a logical timeline of an attack.
 - Write a concise investigative summary suitable for a non-technical audience.
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3. Scenario

On May 15, 2025, at approximately 2:30 PM, Bob, the payroll manager at Innovate Solutions, receives an email that appears to be from the company's IT service provider, "SecureNet." The email warns him that his email account has shown unusual activity and requires him to immediately verify his account by clicking a link. Bob clicks the link, which takes him to a website that looks like his company's login page. He realizes something is wrong and closes the browser. The next day, the real SecureNet team alerts the company of a successful login to the payroll portal from an unrecognized IP address. You have been provided with the email header from the suspicious email Bob received and the firewall logs from the time of the incident.

4. Tools and Resources

A text editor (like Notepad, VS Code, or TextEdit) to view the log files.

An online email header analyzer, such as the one from MXToolbox.

A public cryptocurrency block explorer (e.g., Blockchain.com for Bitcoin).

5. Provided Evidence

Evidence File 1: email_header.txt

```
Return-Path: <bounces@server.suspicious-domain.com>  
Received: from mail-server.suspicious-domain.com (mail-  
server.suspicious-domain.com [198.51.100.24])  
by mail.innovatesolutions.com (Postfix) with ESMTP id 4B9F2C0F4G  
for <bob@innovatesolutions.com>; Tue, 15 May 2025 14:30:15 -0400  
(EDT)  
From: "SecureNet IT Support" <support@securenet.com>  
To: "Bob" <bob@innovatesolutions.com>  
Subject: Urgent: Security Action Required  
Date: Tue, 15 May 2025 14:30:10 -0400 (EDT)
```

...Email Body Content...

Evidence File 2: firewall_log.csv

Timestamp	Source_IP	Destination_IP	Destination_Port	Action
2025-05-15 14:30:25	198.51.100.24	10.0.1.55	443	ALLOW
2025-05-15 14:35:10	203.0.113.112	10.0.0.1	53	ALLOW
2025-05-15 14:40:00	198.51.100.24	10.0.1.70	80	DENY
2025-05-15 15:05:12	203.0.113.112	10.1.10.200	445	DENY
2025-05-16 09:15:45	198.51.100.24	209.10.20.30	443	ALLOW

Evidence File 3: crypto_transaction.txt

On May 17, 2025, a payment of 5 BTC was sent from the attacker's known wallet to a new, unidentified wallet address.

Attacker Wallet (Source): bc1qxy2kgdygjrsqtzq2n0yrf2493p83kkfjhx0wlh

Unidentified Wallet (Destination):
bc1qyl6zlsyqfvjmd3pfs7lwd3rm42q3u0t2d49e8v

Transaction ID:

a1075db55d416d3ca199f55b6084e2115b9345e16c5cf302fc80e9d5fbf5d48d

Note: 10.0.1.55 is the IP address of Bob's workstation. 209.10.20.30 is the public IP address of the company's payroll portal.

6. Step-by-Step Walkthrough

1 Analyze the Email Header:

Copy the contents of email_header.txt.

Go to the MXToolbox Email Header Analyzer and paste the header into the analysis tool.

Examine the results. Pay close attention to the From address versus the actual Received path. The From address is what the user sees, but the Received path shows the true origin of the email.

Identify the sender's real IP address from the Received line. Does it match the From domain?

2 Analyze the Firewall Log:

Open firewall_log.csv in your text editor.

Read the log line by line. Remember the format is
Timestamp,Source_IP,Destination_IP,Destination_Port,Action.

Look for any connections that involve the sender's IP address you identified in Step 1.

Look for any connections to the payroll portal (209.10.20.30). Is there a connection from an IP address that seems suspicious?

3 Correlate the Evidence:

Build a timeline. What happened first? The email or a firewall event?

Did the attacker's IP address (from the email header) connect to Bob's computer? The firewall log has the answer.

The unauthorized login happened on May 16th. Do you see a firewall event that corresponds to this? What is the source IP of that connection?

Based on all the evidence, what is the connection between the phishing email and the unauthorized login a day later?

7. Deliverable: Investigative Summary

Write a brief (150-200 word) investigative summary that explains the incident to the management of Innovate Solutions. Your summary must be clear, concise, and non-technical. It should answer the following questions:

How did the attack start?

What was the attacker's true IP address?

How did the attacker gain access to the payroll portal?

What is the direct evidence linking the initial phishing email to the final unauthorized login?

What is the significance of the cryptocurrency transaction?

8. Assessment Rubric

Criteria	Not Met (0 pts)	Partially Met (5 pts)	Fully Met (10 pts)
Email Analysis	Fails to identify the sender's true IP address.	Correctly identifies the sender's IP but does not explain the discrepancy.	Correctly identifies the sender's IP (198.51.100.24) and explains why it proves the email is a phish.
Firewall Analysis	Fails to identify the relevant log entries.	Identifies one relevant entry but not both.	Correctly identifies both the connection to Bob's PC and the later connection to the payroll portal from the same IP.

Criteria	Not Met (0 pts)	Partially Met (5 pts)	Fully Met (10 pts)
Correlation	Does not connect the email to the final login.	Explains that the same IP was involved but does not form a coherent narrative.	Clearly articulates the full attack chain: phish -> likely credential theft -> fraudulent login from attacker's IP.
Summary/ Clarity	The summary is unclear, inaccurate, or overly technical.	The summary is mostly accurate but is confusing or misses a key point.	The summary is clear, accurate, non-technical, and correctly answers all required questions.

9. Reflection Questions

In this scenario, Bob claimed he didn't enter his credentials. Is it more likely that he is mistaken, or is there another way the attacker could have compromised him just by him clicking the link?

What does the cryptocurrency transaction likely represent in the context of this attack?

What is the single most important preventative measure the company could implement to stop this exact attack from happening again?

10. Additional Resources

[Reading Email Headers - Rackspace](#)

[What is a Firewall Log? - SolarWinds](#)

Instructions for Documenting and Sharing The Project for Peer Review

Document the Project



Use word processing software to create your report.

Ensure all sections of the project document structure are completed.

Proofread and edit your report for clarity and accuracy.



Share the Project

Document: Save the report in PDF format.

Upload: Share in the “Peer Review” area of the course shell with a brief description.



Peer Review

Review the projects submitted by your peers.

Provide constructive feedback and comments on their work.